

Decent Coaching Classes

X Shivraj

Maths Worksheet

Chapter 2: Quadratic Equations

Marks: 60

Section A — Standard Form & Identification (5 × 1 = 5 Marks)

1. Write the following equation in standard form and state the values of a, b, c: $x(x - 2) = 5$ [1]
2. Write the following equation in standard form: $(x + 4)(x - 4) = 2x$ [1]
3. Write the following equation in standard form: $2x^2 = 3(x - 1)$ [1]
4. State whether the equation $x^2 + \frac{1}{x} = 5$ is a quadratic equation. Give a reason. [1]
5. Write the values of a, b, c for the quadratic equation: $x^2 - 7x + 12 = 0$ [1]

Section B — Solving by Factorization (5 × 2 = 10 Marks)

1. Solve by factorization: $x^2 - 5x + 6 = 0$ [2]
2. Solve by factorization: $2x^2 + 7x + 3 = 0$ [2]
3. Solve by factorization: $x^2 - 9x + 20 = 0$ [2]
4. Solve by factorization: $6x^2 - x - 2 = 0$ [2]
5. Solve by factorization: $4x^2 - 4x + 1 = 0$ [2]

Section C — Solving by Completing the Square (3 × 3 = 9 Marks)

1. Solve by completing the square: $x^2 + 6x - 7 = 0$ [3]
2. Solve by completing the square: $2x^2 - 8x - 10 = 0$ [3]
3. Solve by completing the square: $x^2 - 10x + 21 = 0$ [3]

Section D — Solving by Formula Method (4 × 3 = 12 Marks)

1. Solve using the formula method: $3x^2 - 5x + 2 = 0$ [3]
2. Solve using the formula method: $x^2 + x - 12 = 0$ [3]
3. Solve using the formula method: $5x^2 - 6x - 2 = 0$ [3]
4. Solve using the formula method: $2x^2 - 7x + 3 = 0$ [3]

Section E — Nature of Roots (Discriminant) (5 × 2 = 10 Marks)

1. Determine the nature of the roots (without solving): $2x^2 - 5x + 3 = 0$ [2]
2. Determine the nature of the roots (without solving): $x^2 - 4x + 4 = 0$ [2]
3. Determine the nature of the roots (without solving): $3x^2 + 2x + 5 = 0$ [2]
4. Determine the nature of the roots (without solving): $x^2 - 7x + 6 = 0$ [2]
5. Determine the nature of the roots (without solving): $4x^2 - 4x + 1 = 0$ [2]

Section F — Relation Between Roots and Coefficients (3 × 2 = 6 Marks)

1. If α, β are the roots of $2x^2 - 7x + 3 = 0$, find $\alpha + \beta$ and $\alpha\beta$. [2]
2. One root of the equation $x^2 - 5x + k = 0$ is 2. Find the value of k and the other root. [2]
3. Form a quadratic equation whose roots are 4 and -3 . [2]

Section G — Word Problems (2 × 4 = 8 Marks)

1. The sum of the squares of two consecutive positive integers is 313. Find the integers. [4]
2. The product of two consecutive positive multiples of 5 is 300. Find the multiples. [4]

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Answer Key & Solutions

Section A — Standard Form & Identification

- $x(x - 2) = 5 \Rightarrow x^2 - 2x - 5 = 0$; here $a = 1$, $b = -2$, $c = -5$
- $(x + 4)(x - 4) = 2x \Rightarrow x^2 - 16 = 2x \Rightarrow x^2 - 2x - 16 = 0$
- $2x^2 = 3(x - 1) \Rightarrow 2x^2 = 3x - 3 \Rightarrow 2x^2 - 3x + 3 = 0$
- No, it is not a quadratic equation, because the term $\frac{1}{x}$ makes it a rational (non-polynomial) equation — x does not appear with only non-negative integer powers.
- $a = 1$, $b = -7$, $c = 12$

Section B — Solving by Factorization

- $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$
- $2x^2 + 7x + 3 = 0 \Rightarrow (2x + 1)(x + 3) = 0 \Rightarrow x = -\frac{1}{2}$ or $x = -3$
- $x^2 - 9x + 20 = 0 \Rightarrow (x - 4)(x - 5) = 0 \Rightarrow x = 4$ or $x = 5$
- $6x^2 - x - 2 = 0 \Rightarrow (3x - 2)(2x + 1) = 0 \Rightarrow x = \frac{2}{3}$ or $x = -\frac{1}{2}$
- $4x^2 - 4x + 1 = 0 \Rightarrow (2x - 1)^2 = 0 \Rightarrow x = \frac{1}{2}$ (equal roots)

Section C — Solving by Completing the Square

- $x^2 + 6x - 7 = 0 \Rightarrow x^2 + 6x = 7 \Rightarrow x^2 + 6x + 9 = 16 \Rightarrow (x + 3)^2 = 16 \Rightarrow x + 3 = \pm 4 \Rightarrow x = 1$ or $x = -7$
- $2x^2 - 8x - 10 = 0 \Rightarrow x^2 - 4x - 5 = 0 \Rightarrow x^2 - 4x = 5 \Rightarrow x^2 - 4x + 4 = 9 \Rightarrow (x - 2)^2 = 9 \Rightarrow x - 2 = \pm 3 \Rightarrow x = 5$ or $x = -1$
- $x^2 - 10x + 21 = 0 \Rightarrow x^2 - 10x = -21 \Rightarrow x^2 - 10x + 25 = 4 \Rightarrow (x - 5)^2 = 4 \Rightarrow x - 5 = \pm 2 \Rightarrow x = 7$ or $x = 3$

Section D — Solving by Formula Method

- $a = 3$, $b = -5$, $c = 2$; $b^2 - 4ac = 25 - 24 = 1$; $x = \frac{(5 \pm 1)}{6} \Rightarrow x = 1$ or $x = \frac{2}{3}$
- $a = 1$, $b = 1$, $c = -12$; $b^2 - 4ac = 1 + 48 = 49$; $x = \frac{(-1 \pm 7)}{2} \Rightarrow x = 3$ or $x = -4$
- $a = 5$, $b = -6$, $c = -2$; $b^2 - 4ac = 36 + 40 = 76$; $x = \frac{(6 \pm \sqrt{76})}{10} = \frac{(3 \pm \sqrt{19})}{5}$
- $a = 2$, $b = -7$, $c = 3$; $b^2 - 4ac = 49 - 24 = 25$; $x = \frac{(7 \pm 5)}{4} \Rightarrow x = 3$ or $x = \frac{1}{2}$

Section E — Nature of Roots (Discriminant)

- $\Delta = b^2 - 4ac = 25 - 24 = 1 > 0$ and a perfect square $\Rightarrow$ roots are real, rational and unequal
- $\Delta = 16 - 16 = 0 \Rightarrow$ roots are real and equal
- $\Delta = 4 - 60 = -56 < 0 \Rightarrow$ roots are not real (no real roots)
- $\Delta = 49 - 24 = 25 > 0$ and a perfect square $\Rightarrow$ roots are real, rational and unequal
- $\Delta = 16 - 16 = 0 \Rightarrow$ roots are real and equal

Section F — Relation Between Roots and Coefficients

1. $\alpha + \beta = -b/a = 7/2$; $\alpha\beta = c/a = 3/2$

2. Since $x = 2$ is a root: $4 - 10 + k = 0 \Rightarrow k = 6$. Equation becomes $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$, so the other root is 3.

3. Sum of roots = $4 + (-3) = 1$; Product of roots = $4 \times (-3) = -12$. Equation: $x^2 - (\text{sum})x + \text{product} = 0 \Rightarrow x^2 - x - 12 = 0$

Section G — Word Problems

1. Let the integers be x and $x + 1$. Then $x^2 + (x + 1)^2 = 313 \Rightarrow 2x^2 + 2x + 1 = 313 \Rightarrow 2x^2 + 2x - 312 = 0 \Rightarrow x^2 + x - 156 = 0 \Rightarrow (x - 12)(x + 13) = 0 \Rightarrow x = 12$ (rejecting $x = -13$ as the integers are positive). So the integers are 12 and 13.

2. Let the multiples be $5x$ and $5(x + 1)$. Then $5x \times 5(x + 1) = 300 \Rightarrow 25x(x + 1) = 300 \Rightarrow x(x + 1) = 12 \Rightarrow x^2 + x - 12 = 0 \Rightarrow (x + 4)(x - 3) = 0 \Rightarrow x = 3$ (rejecting $x = -4$ as the multiples are positive). So the multiples are 15 and 20.